

PROJECT DOCUMENT — IRIS-3

Version: main-v1.0 | Date: 2025-01-15 | Milestone: CDR

Customer: ESA | Domain: Optical Earth Observation Payload

Decision: GO_WITH_CONSTRAINTS | Confidence: 78%

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1. USER NEEDS

UN-001 [CRITICAL] — The system shall provide ground imagery with sufficient resolution to identify agricultural field boundaries

UN-002 [CRITICAL] — The system shall operate continuously across orbital thermal cycles without performance degradation

UN-003 [CRITICAL] — The payload shall survive launch vibration and shock loads

UN-004 [HIGH] — Imagery data shall be downlinked and processed within acceptable latency

UN-005 [CRITICAL] — The system shall maintain pointing accuracy sufficient for image geolocation

UN-006 [HIGH] — Total payload mass shall fit within platform accommodation envelope

UN-007 [HIGH] — Power consumption shall not exceed platform supply allocation

2. REQUIREMENTS

Domain: OPTICAL

L1-OPT-001 [CRITICAL] | Need: UN-001 | Compliance: 92%

Req: GSD $\leq 1.5\text{m}$ at nadir

Acceptance: GSD measured from calibration target $< 1.5\text{m}$

L1-OPT-002 [CRITICAL] | Need: UN-001 | Compliance: 88%

Req: MTF at Nyquist ≥ 0.40

Acceptance: MTF measured at Nyquist ≥ 0.40

L1-OPT-003 | Need: UN-001 | Compliance: 95%

Req: SNR ≥ 80 dB at reference radiance

Acceptance: SNR ≥ 80 dB under reference illumination

Domain: THERMAL

L1-THM-001 [CRITICAL] | Need: UN-002 | Compliance: 61%

Req: Optical bench $\Delta T \leq 15^\circ\text{C}$ across orbital cycle

Acceptance: Max ΔT during TVT $\leq 15^\circ\text{C}$

L1-THM-002 [CRITICAL] | Need: UN-002 | Compliance: 84%

Req: Detector temperature $200\text{K} \pm 5\text{K}$

Acceptance: Detector temp stability verified in TVT

Domain: MECHANICAL

L1-MEC-001 [CRITICAL] | Need: UN-003 | Compliance: 93%

Req: First natural frequency ≥ 80 Hz

Acceptance: Modal analysis and sine sweep confirm $f_1 \geq 80$ Hz

L1-MEC-002 [CRITICAL] | Need: UN-003 | Compliance: 90%

Req: Withstand random vibration 14.1 Grms (ECSS-E-ST-10-03)

Acceptance: Random vibe test – no structural failure

L1-MEC-003 | Need: UN-003 | Compliance: N/A

Req: Shock response per platform ICD

Acceptance: SRS measured $\leq$ ICD limits at all frequencies

Domain: DATA_PROCESSING

L1-DPR-001 | Need: UN-004 | Compliance: 68%

Req: On-board compression ratio $\geq 4:1$, PSNR ≥ 40 dB

Acceptance: Compression ratio and PSNR verified on reference set

L1-DPR-002 | Need: UN-004 | Compliance: 82%

Req: End-to-end latency $\leq 500\text{ms}$

Acceptance: Latency measured in HIL test $\leq 500\text{ms}$

Domain: OPTICAL

L1-ATT-001 [CRITICAL] | Need: UN-005 | Compliance: 94%

Req: Pointing knowledge error ≤ 0.005° (3σ)

Acceptance: PKE verified by star tracker calibration

Domain: MECHANICAL

L1-MAS-001 | Need: UN-006 | Compliance: 97%

Req: Total mass including harness ≤ 35 kg

Acceptance: Mass measured at AIT ≤ 35 kg

Domain: THERMAL

L1-PWR-001 | Need: UN-007 | Compliance: 91%

Req: Peak power ≤ 180W during imaging mode

Acceptance: Power measured during functional test ≤ 180W

L1-PWR-002 | Need: UN-007 | Compliance: 88%

Req: Average power during eclipse ≤ 45W

Acceptance: Average power during simulated eclipse ≤ 45W

3. SUBSYSTEM STATUS

Subsystem	Score	Status	Detail	Mass (kg)	Volume (m³)
optical	92%	OK	MTF 0.42 > threshold 0.40	8.2	0.018
thermal	61%	WARN	ΔT 14.8°C — margin < 5%	3.1	0.004
mechanical	88%	OK	1st mode 112 Hz > 80 Hz req	12.4	0.031
structure	91%	OK	Mass 33.8 kg / budget 35 kg	6.8	0.022
solar	89%	OK	Power margin +18%	1.8	0.006
startracker	67%	WARN	Stray light analysis pending	0.9	0.001
dataproc	71%	WARN	Compression ratio TBC	1.0	0.002
TOTAL				34.2 kg	

4. V&V; EVIDENCE

Evidence ID	Requirement	Type	Tool	Result	Verdict	Date
SIM-OPT-001	L1-OPT-001	simulation	MATLAB Optical Model	GSD 1.23m	PASS	2024-09-12
TST-OPT-001	L1-OPT-001	test	Lab calibration target	GSD 1.31m	PASS	2024-11-03
SIM-OPT-002	L1-OPT-002	simulation	MATLAB Optical Model	MTF 0.42	PASS	2024-09-15
SIM-OPT-003	L1-OPT-003	simulation	STK Radiometric Model	SNR 84.2 dB	PASS	2024-10-01
TST-OPT-003	L1-OPT-003	test	Integrating sphere	SNR 81.7 dB	PASS	2024-11-20
SIM-THM-001	L1-THM-001	simulation	MATLAB Thermal Model	ΔT 14.8°C	CONDITIONAL	2024-08-20
SIM-THM-002	L1-THM-002	simulation	MATLAB Thermal Model	198.3K ± 3.1K	PASS	2024-08-22
TST-THM-002	L1-THM-002	test	Thermal Vacuum Chamber	199.1K ± 4.2K	PASS	2024-12-05
SIM-MEC-001	L1-MEC-001	simulation	FEM NASTRAN	f1 = 112 Hz	PASS	2024-07-15
TST-MEC-001	L1-MEC-001	test	Vibration test facility	f1 = 108 Hz	PASS	2024-10-28
TST-MEC-002	L1-MEC-002	test	Vibration test facility	14.1 Grms — PASS	PASS	2024-10-30
SIM-DPR-001	L1-DPR-001	simulation	Python pipeline	Ratio 3.8:1 PSNR 39.2dB	CONDITIONAL	2024-11-10
TST-DPR-002	L1-DPR-002	test	HIL test bench	Latency 420ms	PASS	2024-12-01
TST-ATT-001	L1-ATT-001	test	Star tracker calib bench	PKE 0.0028°	PASS	2024-09-30

TST-MAS-001	L1-MAS-001	test	Precision scale AIT	34.2 kg	PASS	2024-12-10
TST-PWR-001	L1-PWR-001	test	Power analyzer AIT	162W peak	PASS	2024-11-25
SIM-PWR-002	L1-PWR-002	simulation	STK Power Budget	38W average	PASS	2024-10-15

5. OPEN NON-CONFORMANCE REPORTS

- NCR-2024-031** [MAJOR] — Thermal gradient margin < 5% on optical bench
Requirement: L1-THM-001 | Owner: Thermal team | Opened: 2024-11-15
- NCR-2024-038** [MINOR] — On-board compression ratio 3.8:1 below 4:1 req
Requirement: L1-DPR-001 | Owner: Data processing team | Opened: 2024-11-12
- NCR-2024-041** [MAJOR] — Shock test not yet performed — evidence missing
Requirement: L1-MEC-003 | Owner: AIT team | Opened: 2024-12-02

6. ECSS OPEN VALIDATION ITEMS

- THM-V-001** | ECSS-E-ST-31C §5.4 | [OPEN]
Activity: Thermal balance test to validate TMM
- THM-V-002** | ECSS-E-ST-31C §5.5 | [OPEN]
Activity: Thermal vacuum test at qualification margins (4 cycles min)
- MEC-V-004** | ECSS-E-ST-32C Rev.1 §9 | [OPEN]
Activity: Shock test per launcher SRS ICD
- STR-V-003** | ECSS-E-ST-32C Rev.1 §10 | [PENDING]
Activity: Acoustic test at qualification level +3dB
- DPR-V-001** | ECSS-E-ST-40C §5.3 | [OPEN]
Activity: On-board compression algorithm validation

7. SYSTEM CONSTRAINTS

- Mass budget: 35.0 kg total | Current: 34.2 kg | Margin: 0.8 kg
- Power budget imaging: 180W max | Current peak: 162W | Margin: 18W
- Thermal ΔT limit: 15°C | Current: 14.8°C | Margin: 0.2°C [CRITICAL]
- First natural frequency: 80 Hz min | Current: 108 Hz | Margin: 28 Hz
- Volume envelope: platform ICD defines max 0.10 m³ | Current: 0.084 m³ | Margin: 0.016 m³